Environ Dev Sustain (2016) 18:187–196
DOI 10.1007/s10668-015-9633-2

Microbial population dynamics under fertigation by distillery effluent in sugarcane-ratoon cropping system

Rajesh Kumar

Received: 26 August 2014 / Accepted: 4 February 2015 / Published online: 11 February 2015
© Springer Science+Business Media Dordrecht 2015

Abstract Distillery effluents, a major by-product of distilleries are a major environmental hazard as they are rich in organic matter, micro- and macroelements. Management of this enormous amount of nutrient-rich effluent and its effective utilization is a major challenge. These effluents can be used as a source of nutrient by C_4 crops like sugarcane and subsequent ratoon crops under fertigation system. Impact of fertigation in sugarcane-ratoon cropping system on soil microbial biomass is important as the microbial population plays an important role in the nutrient and biogeochemical recycling. Keeping this in mind, a field experiment was carried out in the sandy loam soil of semi-arid to humid sub-tropical climate of Gajraula (28.85°N, 78.23°E) region, India with different doses of diluted and undiluted effluent for fertigation of sugarcane crop in presence and absence of 50 % recommended dose of fertilizers with suitable replicates. Two post-sown irrigations with four times diluted effluent gave the maximum yield in case of sugarcane crop. Taking into consideration the yield and the culturable microbial population sizes, it can be concluded that one to two post-sown irrigations with diluted effluent can be used to enhance soil fertility in terms of microbial population and to enhance sugarcane and ratoon crop productivity. This package is economically viable and can also take care of distillery waste management and reduction in consumption of chemical fertilizers by 50 %, a key for sustainable agricultural development.

Keywords BOD · COD · Distillery effluent · Fertigation · Microbial population · Fertilizer

R. Kumar (✉)
Department of Environmental Microbiology, Babasaheb Bhimrao Ambedkar University (A Central University), Lucknow 226025, Uttar Pradesh, India
e-mail: rajesh_skumar@yahoo.co.in

R. Kumar
Department of Microbiology, College of Basic Sciences and Humanities, G. B. Pant University of Agriculture and Technology, Pantnagar 263 145, India

1 Introduction

With more and more industrialization, which is indispensable, the problem of effective effluent management is increasing. The safe disposal of even the treated distillery effluent poses a problem because of the fear that it might render the agricultural land unfit for agricultural purposes. But in case where the effluent is free from toxic heavy metals and other undesirable organo-compounds, utilization of effluent to irrigate agricultural crops has a beneficial effect on soil health and crop yields. It is reported that there are approximately more than 285 distillery industries in India, concentrated mainly in the states of UP, Karnataka and Maharashtra generating approximately 40 billion litres of wastewater annually. Among these, a few of them have adopted the fertigation as the efficient means of wastewater treatment and reuse (Chauhan and Rai 2012) with little success.

Under the present trend of exploitative agriculture, inherent soil fertility can no longer be maintained on a sustainable basis. The nutrient supplying capacity of soil declines steadily under continuous and intensive cropping. Under such circumstances, microbial population can play a major role in the nutrient recycling and energy flow by organic matter decomposition in the soil as well as plant rhizosphere. The wastewater could potentially be used for non-potable uses like industrial processing, landscape irrigation and agricultural irrigation (also known as fertigation/ferti-irrigation). India has many arid and semi-arid regions where the technology of fertigation could be encouraged. If used judiciously, the various micro- and macrocontents of wastewater can act as nutrients for crops thereby reducing the use of additional fertilizers and making it a low-cost option for farmers (Chauhan and Rai 2012). However, the use of effluent may have some effect on the microbial population (which play a major role in nutrient recycling), and therefore, its monitoring is very necessary for designing the fertigation modules. Also the crop to be selected for fertigation module should be more robust. Sugarcane (*Saccharum officinarum* L.), a commercially important C4 plant that produces more amount of biomass/carbonaceous material as compared to C3 plants can be used under fertigation model. It has a worldwide significance as a major source of food (sugar) and by-products (Rath et al. 2012). As per the Food and Agriculture Organization (FAO) statistical database, in India alone, total area harvested in the year 2013 was 5.06 million hectares with a production of 341.2 million metric tonnes.

Irrigation water continues to be the single most important factor dictating the success of crop productivity in arid and semi-arid agro-climatic zones. In the past five decades, the water availability has reduced to half and further reduction is fast approaching. This necessitates using every drop of water that can be recycled back for crop production (Rajkishore and Vignesh 2012). Reuse of wastewater is essential to conserve the limited water resources and to protect the environment. It has now become an option to relieve the demand of fresh water for agriculture, thereby playing an important role in new water utilization patterns (Boyden and Rababah 1996; Butler and MacCornick 1996).

Therefore, the present study was undertaken to utilize treated distillery effluent for fertigation purposes in sugarcane crop, a major crop of the region and to study the impact of different irrigation treatments on the soil microbial flora in presence and absence of recommended dose of fertilizers.

2 Materials and methods

2.1 Distillery effluent characterization

The effluent used in the study was obtained from a distillery located at Gajraula, in the district of Jyotiba Phule Nagar, Uttar Pradesh, India (28.85°N, 78.23°E). It was chemically analysed as per methods described by APHA (Anonymous 2005).

2.2 Description of experimental site and soil analysis

Field experiment was carried out in the same region, i.e. Gajraula (28.85°N, 78.23°E). The climate of the region was semi-arid to humid sub-tropical with sandy loam type of soil, having a pH 6.40 and EC 0.09 dS m^{-1}. The detailed soil properties of the experimental plot were studied before sowing from a depth of 0–30 cm using the Manual for soil analysis by Page et al. (1982).

2.3 Layout of the experiment

Entire experimental plot was divided into 24 sub-plots (size 8 × 3 m with a buffer strip of 1 m in between). Four types of irrigations were planned, I_0—irrigation with plain water, I_1—pre-sown with effluent and rest after sowing with plain water, I_2—one irrigation after sowing with effluent/plain water in the ratio 1:3 at tillering stage of the crop and rest with plain water after sowing, I_3—two irrigations after sowing with effluent/plain water in the ratio 1:4 at tillering and at 30 days after tillering and rest with plain water. Two different levels of fertilizers in these different irrigation patterns were taken, i.e. F_0—no fertilizer, $F_1 = 50$ % of recommended dose of fertilizers, i.e. RDF (50 kg N as urea, 26 kg P as diammonium phosphate and 33 kg K as muriate of potash ha^{-1} as basal dressing and top dressing twice of 50 kg N ha^{-1} at tillering and after 3 months after sowing). The whole experiment was replicated thrice. Ratoon crop received nitrogen in three equal parts after the harvest of sugarcane and then at an interval of 2 months each.

2.4 Sowing of sugarcane crop

Sugarcane was sown in the month of February as per the normal sowing after treatment of the seeds (0.1 % solution of Bavistin for 10 min). Final yield for the crop was taken at the time of harvest. After the harvest of the sugarcane crop, ratoon was left and allowed to grow and observation for yield of sugarcane and ratoon was recorded at the time of harvest. Crop was harvested and dried leaf stripped off from the top most mature internode. The stalks from each plot were tied and weighed to record the yields of millable cane and expressed on per hectare basis.

2.5 Assessment of microbial population

Soil sample for microbial analysis were collected with the help of auger from 0 to 30 cm depth after removing the debris from the sub-surface of the soil. Samples were collected from each of the 24 sub-plots. Each sample was thoroughly mixed before analysis. For assessing microbial population, serial dilutions were prepared using 10 g soil sample from each treatment with three replicates. For assessing total bacterial population, one millilitre

of the serially diluted sample was placed on nutrient agar medium with a pH of 7.0 containing (gL^{-1}) beef extract, 3.0; peptone, 5.0; agar, 15.0. Rose Bengal Agar (Martin, 1950) with streptomycin was used for the isolation of fungi while Kenknight and Munaire's medium containing (g/L^{-1}) dextrose, 1.00; KH_2PO_4, 0.10; $NaNO_3$, 0.10; KCl, 0.10; $MgSO_4{\cdot}7H_2O$, 0.10; and agar, 15.0 was used for actinomycetes isolation. Phosphate solubilizers were isolated on Pikovskaya agar medium containing (g/L^{-1}) dextrose, 10.00; $Ca_3(PO_4)_2$, 5.00; $(NH_4)_2SO_4$, 0.5; $MgSO_4{\cdot}7H_20$, 0.1; KCl, 0.2; yeast extract, 0.5; $MnSO_4{\cdot}H_2O$, 0.0001; $FeSO_4{\cdot}7H_2O$, 0.0001; agar–agar, 15.0; and pH, 7.0 with clearing zone around the colonies. In all the cases, one millilitre of the serially diluted sample was taken over the sterilized medium in petriplates (triplicate) and spread with the help of a sterile glass spreader. Plates were incubated in an inverted position at 30 °C in case of bacteria, actinomycetes and phosphate solubilizers while at 28 °C for fungi. Observations for the total bacterial count were recorded after 24 h, while that for fungi and actinomycetes were recorded after 48 h and phosphate solubilization observations were taken after 96 h.

Same experiment for soil sample collection and microbial population enumeration were repeated again after the harvest of sugarcane crop and then ratoon crop in the subsequent years. Soil samples were again collected from 0 to 30 cm depth after clearing of the leaf debris from the sub-surface and then analysed for the total microbial count as above.

3 Results and discussion

3.1 Effluent characteristics

The physico-chemical characteristics of the treated distillery effluent are shown in Table 1. The distillery effluent was alkaline in nature due to the total dissolved solids and conversion of some elements to their salts after digestion. The distillery effluent was having high biological oxygen demand (BOD), chemical oxygen demand (COD) and was rich in nitrogen (63 mg L^{-1}) and potassium (528 mg L^{-1}) as nutrients (Table 1). Suganya and Rajanan (2009) reported alkaline pH of Post-Methylated Distillery Effluent (PMDE) and also somewhat similar characterization of the effluent samples collected from distillery division of a sugar industry in Tamil Nadu. Likewise by Srivastava et al. (2012), the similar characteristics of effluent were reported from distillery at Gajraula, Uttar Pradesh, India.

Table 1 Chemical composition of distillery effluent

Constituents	Concentration mg L^{-1}
Total N	63
Total K	528
Total Na	212
Total solids	15,980
Total suspended solids	7196
Total dissolved solids	8987
BOD	3825
COD	11,260
EC	2678 μS cm^{-1}
pH	8.3

3.2 Soil characteristics

The experimental field soil was sandy loam in texture having slightly acidic pH (below 7.0) and low EC (0.09 dS m^{-1}) as shown in Table 2. The soil organic carbon was low (0.676 %). The fertility status of field soil revealed 243.3 kg ha^{-1} of available nitrogen while potassium and phosphorus in mid-range. After irrigation with distillery effluent, the soil pH did not changed much and remained slightly acidic. This may be attributed to high buffering capacity of soil and nominal presence of any weak salts. Morlat and Chaussod (2008), Hati et al. (2007) and Srivastava et al. (2012) observed similar results for soil pH, after irrigation with distillery effluent.

The soil microbial population was affected greatly by fertigation with the distillery effluent. All the microbial population like bacteria, actinomycetes and phosphate solubilizing bacteria except fungi showed a gradual increase after irrigation with effluent. The microbial population increased with increasing application of effluent that might be due to greater amount of organic matter substrate (Saliha et al. 2005; Deshpande et al. 2012). Organic material is a natural substrate for saprophytic microorganisms and is essential for formation of soil aggregates. The soil structure ultimately determines the extent of soil aeration and rooting habit of plants. Organic matter also helps in the conservation of soil nutrients by preventing erosion and surface run-off of nutrients. Description of each microbial population is given in the subsequent section.

3.2.1 Impact of fertigation on bacterial population

Bacteria constitute the most abundant group of microorganisms. Both heterotrophic and autotrophic populations can be present in the soil with heterotrophic being the major group. In normal fertile soils, 10^2–10^3 million bacteria are present per g of soil. This number can vary depending on the organic matter content of any particular soil. In the present finding, there was an increase in the total bacterial count over the initial count after harvesting of sugarcane and ratoon crops (Fig. 1). In presence of 50 % of recommended dose of fertilizer, there was an overall increase in the bacterial population as compared to treatments without any fertilizers. Although the chemical fertilizers may result in decrease in microbial population, since the fertilizers were applied at a lower concentration (50 % of RDF) microbial population was not reduced, instead there was an increase. Also, the nutrient-rich nature of the effluent helped bacterial population to develop faster and therefore culturable bacterial population is more in all treatments with effluent irrigated plots over the plain water irrigated plot. Treatments I_2F_1 (three times diluted effluent

Table 2 Characteristics of experimental soil

Parameters	Initial values (before sowing of sugarcane crop)	After harvesting of ratoon crop
pH (1:2) solution	6.40	6.45
EC (1:2) solution	0.090 dS m^{-1}	0.16
Organic carbon (%)	0.676	0.681
Average nitrogen (kg ha^{-1})	243.3	309
Average P_2O_5 (kg ha^{-1})	43.9	Up to 45.3
Average potassium (kg ha^{-1})	188.9	303–1024
Alkaline $KMnO_4$ hydrolysable N (kg ha^{-1})	246	318–345

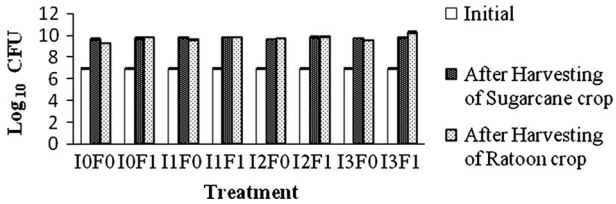

Fig. 1 Total culturable bacterial population initially and after sugarcane and ratoon crop harvesting. CFU, Colony Forming Unit; I_0, Irrigation with plain water; I_1, pre-sown with effluent and rest after sowing with plain water; I_2, one irrigation after sowing, with effluent/plain water in the ratio 1:3 at tillering stage of the crop and rest with plain water after sowing; I_3, two irrigation after sowing with effluent/plain water in the ratio 1:4 at tillering and at 30 days after tillering and rest with pain water; F_0, no fertilizer; F_1, 50 % of recommended dose of fertilizers

irrigation at tillering stage and 50 % recommended dose of fertilizers) and I_3F_1 (two irrigations with four times diluted effluent at tillering and after 30 days of tillering with 50 % RDF) had a profound effect on the bacterial population as compared to initial bacterial counts.

3.2.2 Impact of fertigation on actinomycetes population

The total actinomycetes population is important for the maintenance of fertility of the soil due to the formation of humus, as they feed on complex organic and proteinaceous material and decompose them into simpler compounds. There was a gradual increase in total actinomycetes population in the soil in case of fertigation with distillery effluent and 50 % recommended dose of fertilizer (RDF) as recorded on Kenknight and Munaire's medium over the initial count (Fig. 2).

3.2.3 Impact of fertigation on total fungi

The total fungi count decreased with the time although there was a good content of carbonaceous material in the effluent. This can be concluded that the other microbial strains may have synthesized some antagonistic substances which may have resulted in fungal count reduction and this count gradually decreased with increasing bacterial/actinomycetes/PSB's (phosphate solubilizing bacteria) population (Fig. 3).

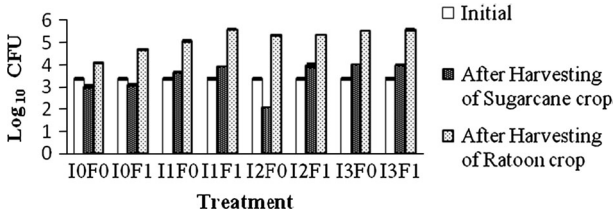

Fig. 2 Total culturable actinomycetes population initially and after sugarcane and ratoon crop harvesting. CFU, Colony Forming Unit; I_0, irrigation with plain water; I_1, pre-sown with effluent and rest after sowing with plain water; I_2, one irrigation after sowing, with effluent/plain water in the ratio 1:3 at tillering stage of the crop and rest with plain water after sowing; I_3, two irrigation after sowing with effluent/plain water in the ratio 1:4 at tillering and at 30 days after tillering and rest with pain water; F_0, no fertilizer; F_1, 50 % of recommended dose of fertilizers

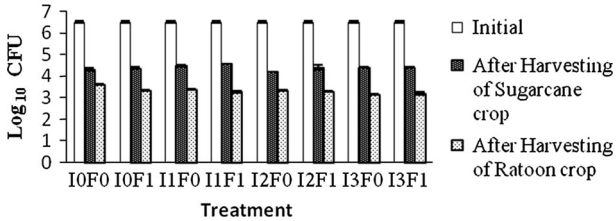

Fig. 3 Total culturable fungal population initially and after sugarcane and ratoon crop harvesting. CFU, Colony Forming Unit; I_0, irrigation with plain water; I_1, pre-sown with effluent and rest after sowing with plain water; I_2, one irrigation after sowing, with effluent/plain water in the ratio 1:3 at tillering stage of the crop and rest with plain water after sowing; I_3, two irrigation after sowing with effluent/plain water in the ratio 1:4 at tillering and at 30 days after tillering and rest with pain water; F_0, no fertilizer; F_1, 50 % of recommended dose of fertilizers

3.2.4 Impact of fertigation on phosphate solubilizing bacteria (PSB's)

Phosphate exists in the soil in the form of insoluble, inorganic salts of phosphate which is not easily available to the plants and microorganisms. Phosphate solubilizers are therefore an important group of organisms in the soil responsible for the synthesis of acids which helps in the solubilization of this non-solubilized form of phosphate that can be easily assimilated by plants and microorganisms for their growth. In the present study, phosphate solubilizing bacteria were monitored and there was an increase in their population in the presence of distillery effluent (Fig. 4). However, in case of I_1 and I_3, the increase was more as compared to I_2, after sugarcane harvesting as well as ratoon harvest. In this case also, 50 % dose of recommended fertilizers had a positive effect on the phosphate solubilizing bacterial population.

When microorganisms grow and multiply on organic matter, carbon is utilized for building the cellular material of microbial cells with the release of carbon dioxide, methane and other volatile substances. In this process, microorganisms also assimilate nitrogen, phosphorous, potassium and sulphur which get bound in the cell protoplasm. Therefore, the C/N, C/P, C/K or C/S ratio in soil are governed by the extent of organic matter utilized by soil microorganisms depending on the oxygen content and the microbial biomass. Thus, three processes go on in the soil simultaneously, i.e. degradation of plant residues, increase in the microbial biomass and liberation or accumulation of end products. The effluent used for irrigating the crop in the present study was rich in these nutrients and played a role in culturable microbial diversity variation.

3.3 Yield of sugarcane and ratoon crop

Agronomic data indicated that fertigation had no impact in improving germination percentage. However, cane yield in case of I_3 was better (approx. 103 t/ha) in presence of 50 % recommended dose of fertilizer than all other irrigation levels (Table 3). Ratoon yield was, however, better in case of I_2 treatment both in the presence and absence of chemical fertilizers, while yield in case of I_3 treatment was slightly less and was 90.12 and 93.08 t/ha in absence and presence of 50 % recommended dose of fertilizer. Microbial population dynamism is dependent on physical, chemical and biological factors. Main crop (sugarcane) and subsequent same crop (ratoon) had a varied microbial population size. Chemical fertilizer helped in increase of plant biomass as well as microbial biomass. It

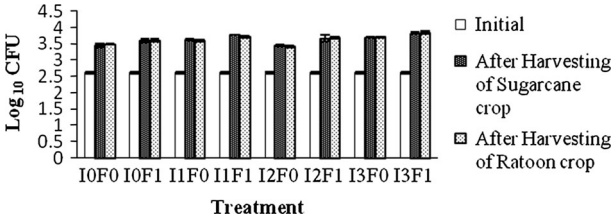

Fig. 4 Total culturable phosphate solubilizing bacterial population initially and after sugarcane and ratoon crop harvesting. CFU, Colony Forming Unit; I_0, irrigation with plain water; I_1, pre-sown with effluent and rest after sowing with plain water; I_2, one irrigation after sowing, with effluent/plain water in the ratio 1:3 at tillering stage of the crop and rest with plain water after sowing; I_3, two irrigation after sowing with effluent/plain water in the ratio 1:4 at tillering and at 30 days after tillering and rest with pain water; F_0, no fertilizer; F_1, 50 % of recommended dose of fertilizers

Table 3 Yield of sugarcane and ratoon crop under different treatments of irrigation with distillery effluent (values are means of three replicates)

Treatment	Millable cane yield (t/ha)	
	Sugarcane crop	Ratoon crop
I_0F_0	81.80 ± 0.124	50.01 ± 0.016
I_0F_1	98.05 ± 0.040	56.15 ± 0.024
I_1F_0	83.21 ± 0.049	80.18 ± 0.029
I_1F_1	97.26 ± 0.020	94.09 ± 0.033
I_2F_0	96.20 ± 0.041	93.52 ± 0.012
I_2F_1	98.31 ± 0.041	98.02 ± 0.008
I_3F_0	99.11 ± 0.028	90.12 ± 0.016
I_3F_1	103.01 ± 0.040	93.08 ± 0.024

where F_0—no fertilizer; F_1—50 % recommended dose of fertilizer; I_0—irrigation with plain water; I_1—pre-sown with effluent and rest after sowing with plain water; I_2—one irrigation after sowing, with effluent/plain water in the ratio 1:3 at tillering stage of the crop and rest with plain water after sowing; I_3—two irrigation after sowing with effluent/plain water in the ratio 1:4 at tillering and at 30 days after tillering and rest with pain water

favoured one population at the cost of other. Correlation between sugarcane and ratoon yield with microbial population size tells that they are positively correlated and increase in the microbial population had a profound effect on crop yield.

Thawale et al. (2011) also reported significant physiological effects on growth of plants in the soil irrigated with effluent from paper mill. Likewise, Chopra et al. (2012) studied the effect of distillery effluent on soil and agronomical characteristics of *Abelmoschus esculentus* L. (okra) and reported enhancement in soil parameters and lowering of pH, but best agronomical growth of okra was seen in 25 % concentration of distillery effluent which is similar to our observation that diluted effluent irrigation is good for plants. Rath et al. (2012) reported that fertigation with 50 % distillery spentwash increased the chlorophyll content and growth in sugarcane as compared to fertilizer and fertilizer + cow dung. Chhaya and Kumar (2014) reported beneficial use of fertigation in agriculture. Effluent irrigation has potential to cut down the fertilizer consumption of the farmers by fifty per cent. Under good soil drainage conditions, application of distillery effluent as pre-sown irrigation with undiluted effluent (I_1) or one irrigation with three times diluted

effluent (I_2) or two irrigations with (1:4) diluted effluent (I_3) helps building up the fertility status of the soil.

4 Conclusion

From the present study, it can be concluded that irrigation with effluents in the post-sown condition is the best as compared to pre-sown irrigation with effluent. Among the various treatments, one to two irrigations with effluent can be used safely to enhance sugarcane and ratoon productivity. Treatment I_3, i.e. two times irrigation, with effluent/plain water in the ratio 1:4 and rest with plain water gave the maximum yield in case of sugarcane crop, while treatment I_2 in case of ratoon crop productivity. The microbial population also varied with the treatment and showed an overall increase except fungal population over the control and pre-sown conditions. Taking into consideration the yield and the culturable microbial population sizes, it can be concluded that one to two post-sown irrigations with diluted effluent can be used to enhance soil fertility in terms of microbial population and to enhance sugarcane and ratoon crop productivity.

This package is not only economically viable but will also solve the problem of treated distillery waste management, increase the final crop yield and maintain soil fertility which is the key for sustainable agricultural development.

Acknowledgments Assistance received from M/s Jubilant Organosys Limited, Gajraula, Uttar Pradesh, India, for providing material support to conduct this study is duly acknowledged.

Conflict of interest The author has declared no conflict of interest.

References

Anonymous, American Public Health Association (APHA). (2005). *Standard methods for the examination of water and wastewater* (21st ed.). Washington, DC: American Public Health Association.

Boyden, B. H., & Rababah, A. A. (1996). Recycling nutrients from municipal wastewater. *Desalination Journal, 106*, 241–246. doi:10.1016/j.desal.2013.12.026.

Butler, R., & MacCornick, T. (1996). Opportunities for decentralized treatment, sewer mining and effluent re-use. *Desalination Journal, 106*, 273–283. doi:10.1016/j.desal.2013.12.026.

Chauhan, J. S., & Rai, J. P. N. (2012). Reuse of distillery wastewater with designed dose and pattern for sugarcane irrigation. *Clean- Soil Air and Water, 40*(8), 838–843.

Chhaya, V., & Kumar, R. (2014). Utilization of distillery wastewater in fertigation: A beneficial use. *International Journal of Research in Chemistry and Environment, 4*(4), 1–9.

Chopra, A. K., Srivastava, S., Kumar, V., & Pathak, C. (2012). Agro-potentiality of distillery effluent on soil and agronomical characteristics of *Abelmoschus esculentus* L. (okra). *Environmental Monitoring and Assessment, 185*, 6635–6645. doi:10.1007/s10661-012-3052-8.

Deshpande, A. N., Kamble, B. M., Shinde, R. B., & Gore, S. B. (2012). Effect of primary treated biomethanated spentwash on soil properties and yield of Sunflower (*Helianthus annuus* L.) on sodic soil. *Communications in Soil Science and Plant Analysis, 43*(4), 730–743. doi:10.1080/00103624.2012.644011.

Food and Agriculture Organisation (FAO) Statistical Database (2014). Accessed November 3, 2014, www.fao.org/statistics/en/

Hati, K. M., Biswas, A. K., Bandyopadhyay, K. K., & Misra, A. K. (2007). Soil properties and crop yields on a vertisol in India with application of distillery effluent. *Soil & Tillage Research, 92*, 60–68. doi:10.1016/j.still.2006.01.011.

Martin, J. P. (1950). Use of acid, rose bengal and streptomycin in the plate method for estimating soil fungi. *Soil Science, 69*(3), 215–232.

Morlat, R., & Chaussod, R. (2008). Long-term additions of organic amendments in a Loire Valley vineyard I. Effects on properties of a calcareous sandy soil. *American Journal of Enology and Viticulture, 59*, 353–363.

Page, A. L., Miller, R. H., & Keeney, D. R. (1982). Total carbon, organic carbon and organic matter. *Methods of Soil Analysis, Part, 2*, 539–579.

Rajkishore, S. K., & Vignesh, N. S. (2012). Distillery spentwash in the context of crop production: A review. *The Bioscan, 7*(3), 369–375.

Rath, P., Pradhan, G., & Misra, M. K. (2012). Effect of distillery spent wash (dsw) and fertilizer on growth and chlorophyll content of sugarcane (*Saccharum officinarum* L.) plant. *Recent Research in Science and Technology, 3*(4), 169–176.

Saliha, B., Kishnakumar, S., Saravanan, A., & Natarajan, S. K. (2005). Microbial and enzyme dynamics in distillery spentwash treated soil. *Research Journal of Agriculture and Biological Sciences, 1*(2), 166–169.

Srivastava, P. C., Singh, R. K., Srivastava, P., & Shrivastava, M. (2012). Utilization of molasses based distillery effluent for fertigation of sugarcane. *Biodegradation, 23*, 897–905. doi:10.1007/s10532-012-9582-5.

Suganya, K., & Rajanan, G. (2009). Effect of one time post-sown and pre-sown application of distillery spentwash on the growth and yield of maize crop. *Botany Research International, 2*(4), 288–294.

Thawale, P. R., Yadav, S. K., Wanjari, T., Singh, S., Juwarkar, A., & Kumar, R. (2011). Monitoring colour and COD removal capacity of soil and assessment of growth performance of crop grown with pulp and paper mill waste water: A lysimeter study. *International Journal of Environmental Engineering, 3*(2), 146–163.